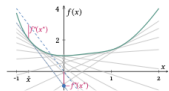
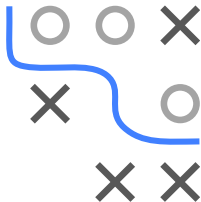


Optimization in Machine Learning

Constrained Optimization

Other forms of duality



(a) The function f (solid) and the linear function of $x^* = -4$ (dashed) and its shifted tangent (dotted).



(b) The Fenchel conjugate f^* of f .

Learning goals

- Dual norms
- Conjugate functions
- Fenchel duality
- Examples in statistics

CONSTRAINED MINIMIZATION AND DUAL NORMS

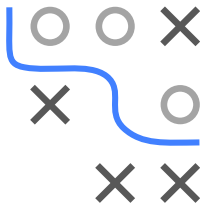
- Consider norm minimization under linear constraints in its primal form:

$$\min_{\mathbf{x} \in \mathbb{R}^d} \|\mathbf{x}\| \quad \text{s.t. } \mathbf{G}\mathbf{x} = \mathbf{h}$$

- $\|\cdot\|$ is some norm function
- If the norm is the L_1 norm, this is the famous basis pursuit problem

► [Click for source](#)

- Question:** Is there a more straightforward way to solve constrained optimization problems involving norms?



CONSTRAINED MINIMIZATION AND DUAL NORMS

- The concept of the **dual norm** from functional analysis can help

Definition Let $\|\mathbf{x}\|$ be the norm of $\mathbf{x}$. The dual norm $\|\mathbf{x}\|_*$ is defined as

$$\|\mathbf{x}\|_* = \max_{\|\mathbf{z}\| \leq 1} \mathbf{z}^T \mathbf{x}$$

- If $\|\mathbf{x}\|$ is a norm and $\|\mathbf{x}\|_*$ its dual norm, then

$$|\mathbf{z}^T \mathbf{x}| \leq \|\mathbf{z}\| \|\mathbf{x}\|_*$$

Example The dual norm of the L_p norm $\|\cdot\|_p$ is the L_q norm $\|\cdot\|_q$ where $1/p + 1/q = 1$

